Case Study Assignment: Warehouse and Retail Sales Analysis

Part 1: Data Preparation

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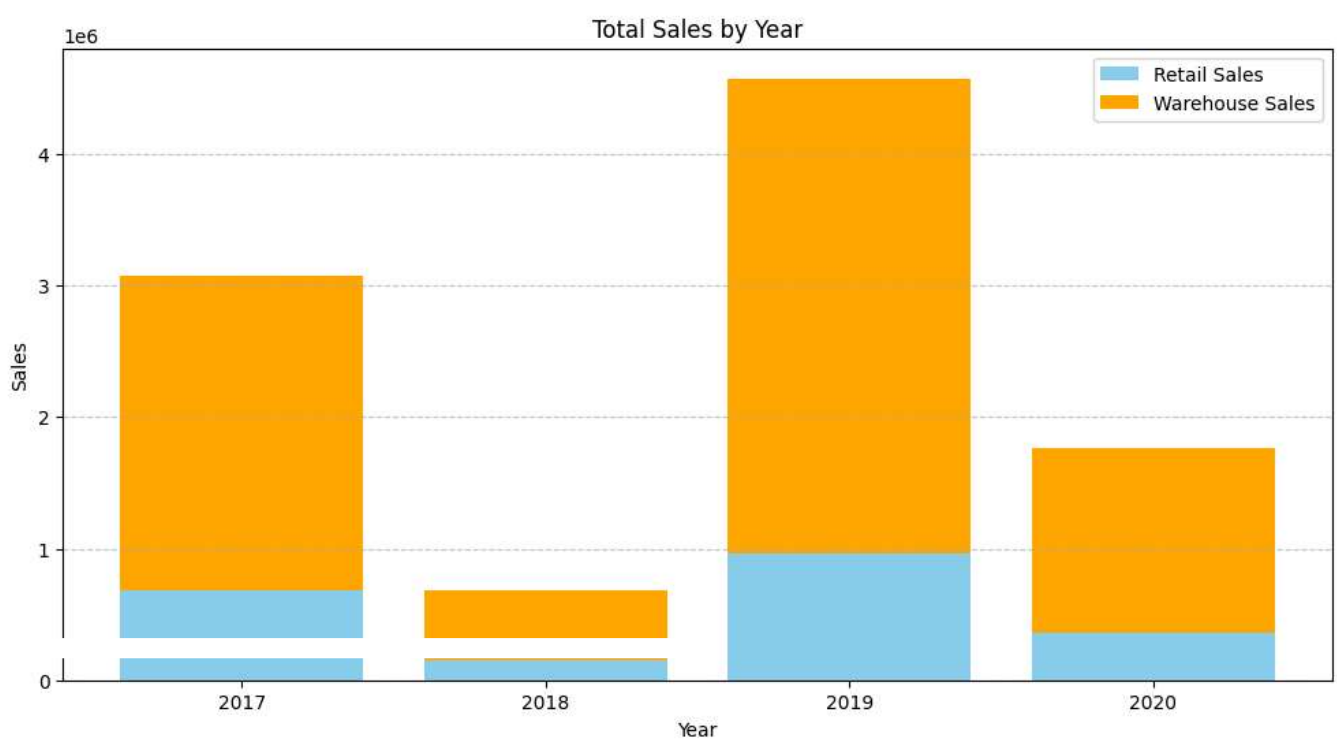
Part 2: Exploratory Data Analysis

```
# Calculate total sales by adding RETAIL SALES and WAREHOUSE SALES year wise


total_sales_per_year = df_cleaned.groupby('YEAR')[['RETAIL SALES', 'WAREHOUSE SALES']].sum()

total_sales_per_year['TOTAL SALES'] = total_sales_per_year['RETAIL SALES'] + total_sales_per

print(total_sales_per_year)
```

```
       RETAIL SALES  WAREHOUSE SALES  TOTAL SALES
YEAR
2017     686760.02       2382157.17   3068917.19
2018     153598.36        530177.51    683775.87
2019     960205.14       3613686.57   4573891.71
2020     360379.22       1399486.62   1759865.84
```

```
# Visualzie total sales by adding RETAIL SALES and WAREHOUSE SALES year wise in Lakhs


import matplotlib.pyplot as plt

total_sales_per_year = df_cleaned.groupby('YEAR')[['RETAIL SALES', 'WAREHOUSE SALES']].sum()
total_sales_per_year['TOTAL SALES'] = total_sales_per_year['RETAIL SALES'] + total_sales_per

total_sales_per_year = total_sales_per_year.reset_index()

plt.figure(figsize=(12, 6))

plt.bar(total_sales_per_year['YEAR'], total_sales_per_year['RETAIL SALES'], label='Retail Sa
plt.bar(total_sales_per_year['YEAR'], total_sales_per_year['WAREHOUSE SALES'], bottom=total_

plt.title('Total Sales by Year')
plt.xlabel('Year')
```

```
plt.ylabel('Sales')
plt.xticks(total_sales_per_year['YEAR'])
plt.legend()
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```




```
# Top 5 suppliers based on total sales


df_cleaned['TOTAL SALES'] = df_cleaned['RETAIL SALES'] + df_cleaned['WAREHOUSE SALES']
total_sales_by_supplier = df_cleaned.groupby('SUPPLIER')['TOTAL SALES'].sum()

top_suppliers = total_sales_by_supplier.sort_values(ascending=False).head(5)

print(top_suppliers)
```

```
SUPPLIER
CROWN IMPORTS             1736309.17
MILLER BREWING COMPANY    1512604.65
```

```
ANHEUSER BUSCH INC        1509222.07
HEINEKEN USA               886035.37
E & J GALLO WINERY         363673.65
Name: TOTAL SALES, dtype: float64
<ipython-input-37-df11f46c190b>:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user
  df_cleaned['TOTAL SALES'] = df_cleaned['RETAIL SALES'] + df_cleaned['WAREHOUSE SALES']
```

```
# Top 10 best-selling items (based on total sales)

df_cleaned['TOTAL SALES'] = df_cleaned['RETAIL SALES'] + df_cleaned['WAREHOUSE SALES']
total_sales_per_item = df_cleaned.groupby('ITEM CODE')[['TOTAL SALES']].sum()

total_sales_per_item = total_sales_per_item.sort_values(by='TOTAL SALES', ascending=False)

top_10_items = total_sales_per_item.head(10)

top_10_items_details = df_cleaned[df_cleaned['ITEM CODE'].isin(top_10_items.index)][['ITEM C

top_10_items_with_details = pd.merge(top_10_items, top_10_items_details, on='ITEM CODE')

top_10_items_with_details = top_10_items_with_details.reset_index(drop=True)
top_10_items_with_details.index += 1

print(top_10_items_with_details)
```

```
   ITEM CODE  TOTAL SALES                    ITEM DESCRIPTION ITEM TYPE
1      23445    328224.83        CORONA EXTRA LOOSE NR - 12OZ      BEER
2      96750    257679.62         CORONA EXTRA 2/12 NR - 12OZ      BEER
3      23886    189711.21            HEINEKEN LOOSE NR - 12OZ      BEER
4      96970    162224.15             HEINEKEN 2/12 NR - 12OZ      BEER
5      90590    148926.40         MILLER LITE 30PK CAN - 12OZ      BEER
6      96741    132665.61          CORONA EXTRA 4/6 NR - 12OZ      BEER
7      25883    121846.80  MODELO ESPECIAL 24 LOOSE NR - 12OZ      BEER
8      90468    108615.97                  BUD LIGHT 30PK CAN      BEER
9      96083    105184.92              HEINEKEN 4/6 NR - 12OZ      BEER
10     23314    100664.44         CORONA EXTRA 18PK NR - 12OZ      BEER
<ipython-input-47-53166d2309ce>:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user
  df_cleaned['TOTAL SALES'] = df_cleaned['RETAIL SALES'] + df_cleaned['WAREHOUSE SALES']
```

Calculated the total sales by summing RETAIL SALES and WAREHOUSE SALES on a yearly basis. This provided a clearer view of sales trends over the years. I then visualized the total sales in lakhs to identify yearly patterns more effectively.

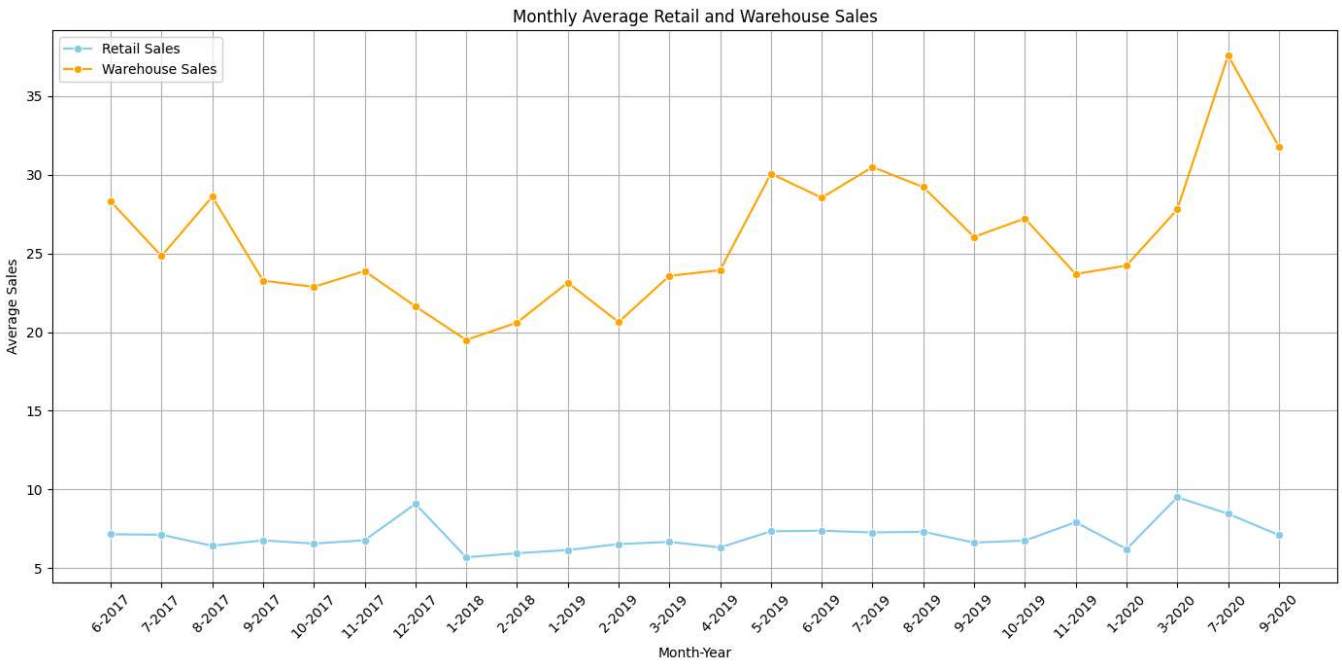
Next, Identified the top 5 suppliers by total sales, with suppliers like CROWN IMPORTS, MILLER BREWING COMPANY AND ANHEUSER BUSCH INC leading in sales volume.

Additionally, analyzed the top 10 best-selling items based on total sales, revealing that popular beer brands such as CORONA EXTRA and HEINEKEN dominated the market.

Start coding or generate with AI.

PART 3 : Business Analysis

```
# Monthly average retail sales and warehouse sales by Month-Year'

import seaborn as sns

monthly_avg_sales = df_cleaned.groupby(['YEAR', 'MONTH'])[['RETAIL SALES', 'WAREHOUSE SALES'

monthly_avg_sales['Month-Year'] = monthly_avg_sales['MONTH'].astype(str) + '-' + monthly_avg

plt.figure(figsize=(14, 7))
sns.lineplot(data=monthly_avg_sales, x='Month-Year', y='RETAIL SALES', marker='o', label='Re
sns.lineplot(data=monthly_avg_sales, x='Month-Year', y='WAREHOUSE SALES', marker='o', label=

plt.title('Monthly Average Retail and Warehouse Sales')
plt.xlabel('Month-Year')
plt.ylabel('Average Sales')
plt.xticks(rotation=45)
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```




```
# Total sales for each item

total_sales_per_item_type = df_cleaned.groupby('ITEM TYPE')[['RETAIL SALES', 'WAREHOUSE SALE
total_sales_per_item_type['TOTAL SALES'] = total_sales_per_item_type['RETAIL SALES'] + total

total_sales_per_item_type = total_sales_per_item_type.sort_values(by='TOTAL SALES', ascendir

print(total_sales_per_item_type)

top_item_type = total_sales_per_item_type.iloc[0]
print(f"The item type that contributes the most to overall sales is: {top_item_type.name}")
```

```
               RETAIL SALES  WAREHOUSE SALES  TOTAL SALES
ITEM TYPE
BEER             574220.86       6527995.14   7102216.00
WINE             746526.31       1157428.28   1903954.59
LIQUOR           802706.59         95004.19    897710.78
KEGS                  0.00        118623.00    118623.00
NON-ALCOHOL       34084.39         26159.26     60243.65
STR_SUPPLIES       2740.88             0.00      2740.88
REF                 663.71           298.00       961.71
DUNNAGE               0.00             0.00         0.00
The item type that contributes the most to overall sales is: BEER
```

Calculated the monthly average sales for each year by aggregating RETAIL SALES and WAREHOUSE SALES. A 'Month-Year' column was created to facilitate better plotting and visualization of monthly trends for both retail and warehouse sales, providing insights into seasonality and monthly sales fluctuations.

Analyzed total sales by item type by summing both retail and warehouse sales across item categories. This analysis revealed that BEER had the highest total sales, followed by WINE and LIQUOR, indicating that alcoholic beverages contribute the most to overall sales.

Analyse whether there are any seasonal trends in sales data ?

Yes there are seasonal trends -By calculating the monthly average sales for each year, the data can reveal fluctuations that suggest seasonality. Certain months consistently lie JULY show higher average sales in 2019 and 2020, particularly around holidays, like BEER, can be linked to seasonality. As it is the Highest selling product

Part 4 : Recommendations and Insights

Insights :

1. Warehouse Sales Outperform Retail:
   The data shows that warehouse sales are more successful than retail sales. They should invest more in digital marketing and advertising to boost the online presence and sales.
2. Strong Performance in 2019/20:
   The warehouse had a great year in 2019/20. They need to explore customer behavior during that time, especially looking at the impact of new product launches and seasonal trends that may have contributed to this success.
3. High Demand for Beer:
   Beer is the top-selling product. They should create targeted marketing strategies for beer

customers, offering exclusive promotions, discounts, and special collections. It's also important to manage stock levels effectively to prevent any out-of-stock situations.

4. Monitor Competitor Activity:
   Keep an eye on competitors' strategies especially when the sales are less. This will help to stay competitive and identify new opportunities in the market.

Recommendations:

1. Expand digital marketing efforts to enhance online visibility and reach a wider audience.
2. Strengthen the e-commerce platform for a smoother and more user-friendly shopping experience.
3. Conduct competitor analysis and follow market trends to prioritize investments effectively.
4. Use customer segmentation to target specific groups with tailored marketing strategies.
5. Incorporate feedback and offer personalized promotions to improve customer engagement.
6. Implement a loyalty program for frequent customers, possibly using a points-based system.
7. Identify causes of retail sales decline and improve the in-store experience, while integrating online and offline channels, like click-and-collect or in-store discounts.
8. Track leads who are aware of the company and send them personalized offers and incentives to encourage their first purchase.

Technical Recommendations:

1. Can use Google Data Studio which is free and easy to integrate with Google Analytics and other tools to visualize data.

2. Customer Relationship Management (CRM) Tools for customer engagement, and sales tracking. Also can use zoho Inventory for managing orders, warehouses,etc.

3. DevOps Implementation:

   GitHUB for version control and managing deployments efficiently.

   Docker: Container platform for creating lightweight, portable applications that can run consistently across various environments.

   Jenkins: Automating software builds, tests, and deployments, perfect for implementing CI/CD pipelines at a low cost.

4. Invest in modern UX/UI design practices to create a seamless shopping experience. Use A/B testing to continuously improve website layout, reducing friction in the customer journey. Prioritize responsive and mobile-first design to cater to the increasing number of mobile shoppers

Double-click (or enter) to edit